

Analysis of Kyoto Emissions Post Mid-Century

The analysis focuses on the emissions profile, particularly those related to the Kyoto gases, post mid-century. The data indicates a significant increase in total emissions from 2050 to 2070, rising from approximately 1560.46 Mt to 3006.96 Mt. This represents a growth rate of 92.7% over the 20-year period.

Emissions Trends

- **Kyoto Gases:** These gases, including CO₂, CH₄, N₂O, and F-gases, show a consistent increase in emissions over the years. The total Kyoto gases emissions rise from 780.23 Mt in 2050 to 1463.27 Mt in 2070, marking an 87.5% increase.
- **CO₂ Emissions:** CO₂ constitutes the largest share of Kyoto gases emissions, with values increasing from 687.51 Mt in 2050 to 1317.99 Mt in 2070. This represents a 92% increase.
- **Non-CO₂ Gases:** The share of non-CO₂ Kyoto gases (CH₄, N₂O, F-gases) also increases. Notably, N₂O emissions rise significantly from 66.77 Mt in 2050 to 170.23 Mt in 2070.

Emissions Shares

- **CO₂ Share:** CO₂'s share of total Kyoto gases emissions remains relatively stable, slightly decreasing from 44.06% in 2050 to 43.83% in 2070.
- **Non-CO₂ Gases Shares:**
 - **CH₄** decreases from 0.14% in 2050 to 0.07% in 2070.
 - **N₂O** increases from 4.28% to 5.66% over the same period.
 - **F-gases** show a slight increase in share from 1.04% to 1.41%.

Key Insights

- The overall increase in emissions post mid-century is driven primarily by CO₂ and N₂O.
- While CO₂ remains the dominant gas in terms of emissions volume, N₂O exhibits a significant growth rate.
- The stability in CO₂'s share of total emissions, alongside increases in other gases, suggests a diversification in the sources of emissions.

Summary

- Total emissions more than double from 2050 to 2070.
- CO₂ emissions increase by 92%, remaining the largest contributor.
- N₂O shows the most significant relative increase among all gases.
- Emissions shares indicate a slight decrease in CO₂'s dominance, with increases in other gases, particularly N₂O and F-gases.

Kyoto GHG Emissions

